Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/25/2013
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	301-405-3993
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City:	College Park	**State:**	MD
Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
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Flight Program:			
Flight Assignment:	NOTE: End date extended to 12/31/2014 per NSSC information (Ed., 9/17/13) NOTE: End date extended to 12/31/2013 per PI (Ed., 10/12/2012)		
Key Personnel Changes/Previous PI:	Not applicable.		
COI Name (Institution):	Sunderland, Peter (University of Maryland)		
Grant/Contract No.:	NNX10AD98G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A novel experimental apparatus, the Burning Rate Emulator (BRE), is being developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of 'virtual' solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it will be possible to emulate steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements will be conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station. The Burning Rate Emulator (BRE) will simulate the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE will control the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This will allow it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment will be considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) will be considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows. 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	• A successful Requirements Definition Review (RDR) was held in Cleveland in June, 2013. • The Science Requirements Document (SRD) for BRE was finalized. • The BRE portions of the ACME integrated SRD were finalized. • Tests were performed to evaluate the ability of BRE to emulate condensed fuels using gaseous fuels in normal gravity. These tests were very successful. • Tests were performed to calibrate the heat flux gages and to better understand the effects on their output of the cooling water conditions. • New burners for spaceflight were designed. Two similar burners were fabricated at UMD. • Analytical models of the flames and burners were further developed. • Yi Zhang successfully defended his M.S. thesis in Fire Protection Engineering and will start the Ph.D. program in Mechanical Engineering in January. • Matt Kim started his M.S. degree in Fire Protection Engineering. • One journal paper was accepted. • Four conference papers were published and presented.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Abstracts for Journals and Proceedings	Zhang Y, Bustamante MJ, Sunderland PB, Quintiere JG, Ferkul P. "A Burning Rate Emulator for Microgravity Fire Safety." Seventh International Seminar on Fire and Explosion Hazards, Providence, Rhode Island, May 5-10, 2013. Poster. Seventh International Seminar on Fire and Explosion Hazards, Providence, Rhode Island, May 5-10, 2013. , May-2013
Articles in Peer-reviewed Journals	Zhang Y, Bustamante MJ, Gollner MJ, Sunderland PB, Quintiere JG. "Burning on flat wicks at various orientations." Journal of Fire Sciences. First published on August 13, 2013 as http://dx.doi.org/10.1177/0734904113495650 , Aug-2013
Papers from Meeting Proceedings	Bustamante MJ, Zhang Y, Dotson KT, Sunderland PB, Quintiere JG. "Burning on Flat Wicks at Various Orientations." Seventh International Seminar on Fire and Explosion Hazards, Providence, Rhode Island, May 5-10, 2013. In: Fire and Explosion Hazards. Proceedings of the Seventh International Seminar on Fire and Explosion Hazards. Edited by D. Bradley, G. Makhviladze, V. Molkov, P. Sunderland and F. Tamanini. Singapore : Research Publishing, 2013. 10 p. http://dx.doi.org/10.3850/978-981-07-5936-0_03-02 , May-2013
Papers from Meeting Proceedings	Zhang Y, Bustamante MJ, Dotson KT, Sunderland PB, Quintiere JG. "A Gas Burner to Emulate Condensed-Phase Burning." Seventh International Seminar on Fire and Explosion Hazards, Providence, Rhode Island, May 5-10, 2013. In: Fire and Explosion Hazards. Proceedings of the Seventh International Seminar on Fire and Explosion Hazards. Edited by D. Bradley, G. Makhviladze, V. Molkov, P. Sunderland and F. Tamanini. Singapore : Research Publishing, 2013. 9 p. http://dx.doi.org/10.3850/978-981-07-5936-0_03-01 , May-2013

Papers from Meeting Proceedings	Zhang Y, Bustamante MJ, Sunderland PB, Quintiere JG, Ferkul P. "A Burning Rate Emulator for Study in Microgravity." 8th U.S. National Combustion Meeting, Organized by the Western States Section of the Combustion Institute, Park City, Utah, May 19-22, 2013. 8th U.S. National Combustion Meeting, Organized by the Western States Section of the Combustion Institute, Park City, Utah, May 19-22, 2013. 14 p. , May-2013